

Lecture 16

Rotor loads, Part 2

ASDA/Dynamics of Rotor Systems
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This lecture

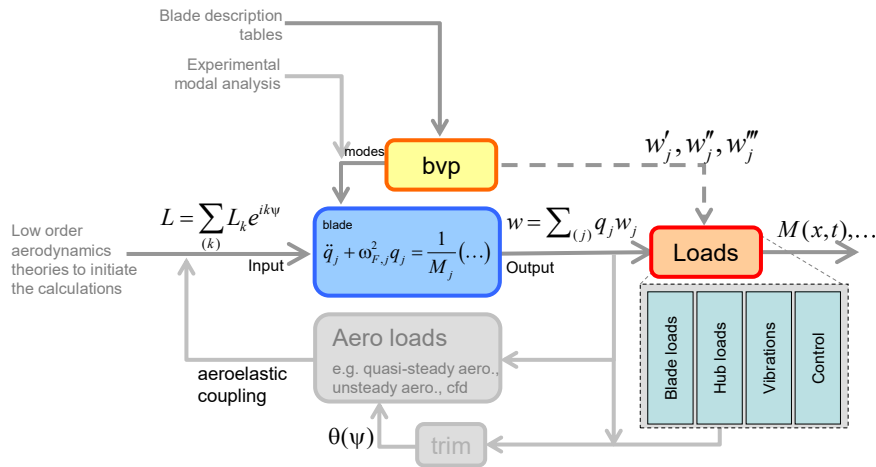
- Expanded load prediction framework
- Introduction to rotor hub loads
- Load transformation rules and hub filtering effect

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Expanded load calculation framework

Expanded rotor loads prediction architecture developed so far:



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Blade load summary

Principal F and L dynamics of rotating blades:

$$(EI_F w'')'' - (T w')' + m\ddot{w} = L(x, t)$$

$$(EI_L v'')'' - (T v')' + m(\ddot{v} - \Omega^2 v) = D(x, t)$$

Blade dynamics using modal coordinates q ($j=1, \dots, N_{M,F}$; $i=1, \dots, N_{M,L}$):

$$\ddot{q}_j + (\omega_{0,F,j}^2 + K_{F,j} \Omega^2) q_j = L_j / M_{F,j}$$

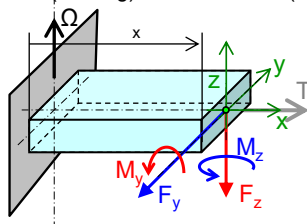
$$\ddot{q}_i + (\omega_{0,L,i}^2 + (K_{L,i} - 1) \Omega^2) q_i = D_i / M_{L,i}$$

Blade response using modal synthesis for $N_{M,F} + N_{M,L}$ modes:

$$w(x, t) = \sum_{j=1}^{N_{M,F}} w_j(x) q_j(t)$$

$$v(x, t) = \sum_{i=1}^{N_{M,L}} v_i(x) q_i(t)$$

Blade (rotating) sectional loads (differential method):



$$F_y = T v'(x, t) - (EI_L v''(x, t))'$$

$$F_z = T w'(x, t) - (EI_F w''(x, t))'$$

$$M_y = EI_F w''(x, t)$$

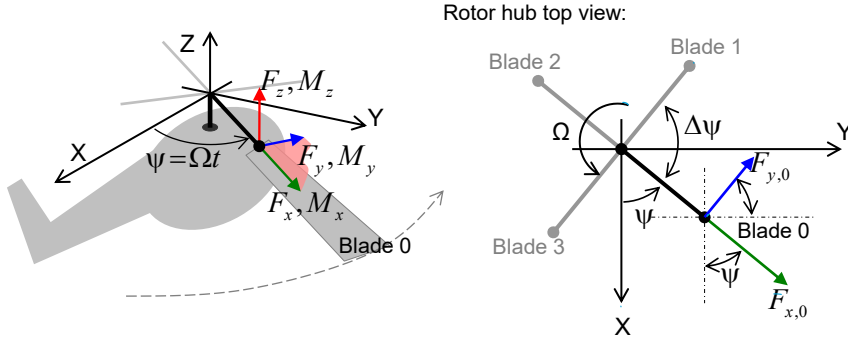
$$M_z = EI_L v''(x, t)$$

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Introduction to rotor hub loads

Blade loads are transmitted to the hub via blade root or hinge system, pitch links, lag dampers, etc. These loads are periodic functions of time ($T=2\pi/\Omega$) during steady flight. Apart from the phase difference, all blades are loaded with the same loading patterns during such regimes. The azimuth angle of the k -th blade: $\psi_k = \psi + \Delta\psi \times b$, $\Delta\psi = 2\pi/N_B$, $\psi = \Omega t$.

The rotating blade root loads need to be transformed to the non-rotating (hub) system (hub) and added together to give the total nonrotating hub loads.



Loads due to blade k in the hub frame:

$$X_k = F_{x,k} \cos \psi_k - F_{y,k} \sin \psi_k, \quad Y_k = F_{x,k} \sin \psi_k + F_{y,k} \cos \psi_k, \quad Z_k = F_{z,k}$$

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Introduction to rotor hub loads, cont.

A sketch of the analytical derivation, or the numerical computational sequence used to obtain the non-rotating rotor hub loads:

Step 1: The steady rotating blade loads are expressed using *Fourier series*

$$F_{x,k} = P_0 + \sum_{(j)} P_j \cos(j\psi_k) + Q_j \sin(j\psi_k)$$

$$F_{y,k} = S_0 + \sum_{(j)} S_j \cos(j\psi_k) + T_j \sin(j\psi_k)$$

$$F_{z,k} = U_0 + \sum_{(j)} U_j \cos(j\psi_k) + V_j \sin(j\psi_k)$$

Substitution 1

Step 2: Blade loads from the blade k in the non-rotating hub frame

$$X_k = F_{x,k} \cos \psi_k - F_{y,k} \sin \psi_k$$

$$Y_k = F_{x,k} \sin \psi_k + F_{y,k} \cos \psi_k$$

$$Z_k = F_{z,k}$$

Substitution 2

Step 3: The total non-rotating hub loads from the N_B blades

$$X = \sum_{k=0}^{N_B-1} X_k$$

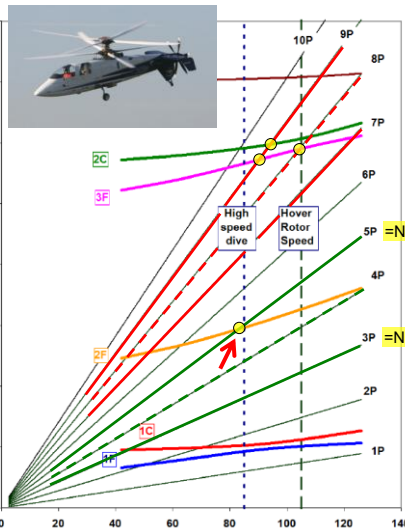
$$Y = \sum_{k=0}^{N_B-1} Y_k$$

$$Z = \sum_{k=0}^{N_B-1} Z_k$$

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Example application

This case illustrates a combined application of frequency diagram information and insights from rotor hub load transformation rules in early design and development.



D. Walsh, S. Weiner, K. Arifian, T. Lawrence, M. Wilson, T. Millott and R. Blackwell, *High Airspeed Testing of the Sikorsky X2 Technology™ Demonstrator*, Presented at the 67th Annual Forum of the American Helicopter Society International, Virginia Beach, VA, May 4, 2011

Dynamics, Design, p.8

The predecessor XH-59A ... had three blades per rotor and no vibration control system experienced vibration well above desirable levels. ... high levels of 2/rev blade root flap moments were the dominant sources of the 3/rev vibration. ... the decision was made to use four-blade rotors.

Dynamics, Blade Natural Frequencies ..., p.8

As a stiff-inplane design ... Damping ratios remained in the 1 – 3 percent of critical damping range throughout the flight envelope.

Dynamics, 4/rev hub loads, p.8

Examination showed that the 4/rev blade root moments are higher than the 3 and 5/rev moments, but none of the loads build up dramatically with airspeed. By comparison, the 2/rev root flap moments, which would contribute to 3/rev vibration on a 3-bladed rotor, were significantly larger than the 3/rev moments so the decision to use four rather than three blades per rotor appears to have been the correct one.

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Summary

- The developed load-calculation framework combines modal analysis (BVP) with transient or steady-state blade response calculations in the time or azimuth domain. Additional assumptions (e.g., rotor symmetry and steady flight conditions) allow the calculation of the overall non-rotating hub loads.
- Blade-passing frequencies are a key vibration characteristic in rotary-wing fuselage vibrations: $jN_B\Omega$, $j=1,2,\dots$
- Rotor filtering rules permit further causal linking between the nonrotating hub and rotating blade loads, e.g.:
 - in-plane rotating vectors: $(j\pm1)\Omega$, $j=1,2,\dots$
 - out-of-plane rotating vectors: $j\Omega$, $j=1,2,\dots$
- Further development of the framework for use in rotor aero-structural performance calculations could involve direct coupling of the existing structure with, at a basic level, quasi-steady and prescribed rotor inflow aerodynamics.



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